

Activity-4 :

Model Driven Development for a banking system

- (1) Model Driven Development (MDD) is a software development approach where software models are created first and application code is generated automatically from those models.

For a banking system:

create UML models such as class diagrams & use case diagrams.

Define customers, accounts, transactions & loans in the models

Transform the models into source code automatically.

This reduces coding errors and development time

(2)

(PIM)

Describes system functionally without technology details

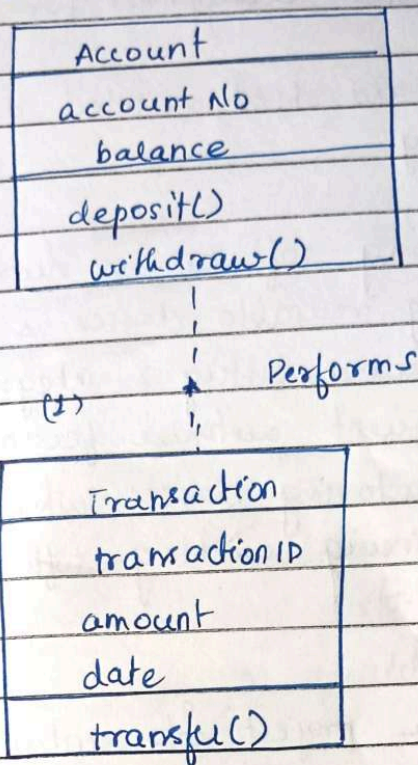
UML class diagram
Use case diagram
Banking business rules
customer, account,
Transaction

(PSM)

Describes implementation using a specific platform

Java classes.
Spring Boot Application
MySQL Database Tables.
Java + MySQL, REST
API

3. UML class diagram



(4) Advantages of generating code from models.

- ⇒ Reduces coding errors.
- ⇒ Faster software development
- ⇒ Improves software quality
- ⇒ Easier maintenance
- ⇒ Consistent code generation
- ⇒ Better documentation.
- ⇒ Increases developer productivity.

(5) Limitations of using MDD in large scale projects

- Requires skilled model designers
- Initial setup cost is high.
- Generated code may require manual modification
- Complex models are difficult to manage.
- Tool compatibility issues
- Performance optimization may be limited.